HW5

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Progress

- Add Preface/Known results to top of each problem/section
- General checks

Q1: 2.15 (c)

Consider the probability space $\big((0,1), \mathcal{B}((0,1)), m\big)$, where $m(\cdot)$ is the Lebesgue measure.

Let $F : \mathbb{R} \to \mathbb{R}$ be a cdf. For $0 < x < 1$, let

$$F_1^{-1}(x) = \inf\{y : y \in \mathbb{R},\ F(y) \geq x\}$$
$$F_2^{-1}(x) = \sup\{y : y \in \mathbb{R},\ F(y) \leq x\}.$$

Let Z_i be the random variable defined by

$$Z_i = F_i^{-1}(x), \quad 0 < x < 1, \quad i = 1, 2.$$

(i)

Find the cdf of Z_i, $i = 1, 2$.

(Hint: Verify using the right continuity of F that for any $0 < x < 1$, $t \in \mathbb{R}$, $F(t) \geq x \iff F_1^{-1}(x) \leq t$.)

Answer

For $x \in (0,1)$, let $A_x \equiv \{y \in \mathbb{R} : F(y) \geq x\}$ and $B_x \equiv \{y \in \mathbb{R} : F(y) \leq x\}$, so that $F_1^{-1}(x) = \inf A_x$ and $F_2^{-1}(x) = \sup B_x$. Since F is nondecreasing with $F(y) \to 0 < x$ as $y \to -\infty$ and $F(y) \to 1 > x$ as $y \to \infty$ (Definition 2.2.4), A_x is nonempty and bounded below, and B_x is nonempty and bounded above. Thus $F_1^{-1}(x)$ and $F_2^{-1}(x)$ are finite.

Proof. **Claim.** For any $x \in (0,1)$ and $t \in \mathbb{R}$,

$$F(t) \geq x \iff F_1^{-1}(x) \leq t. \tag{1}$$

Let $x \in (0,1)$ and $t \in \mathbb{R}$ be given, and let $q \equiv F_1^{-1}(x)$.

($\Rightarrow$) If $F(t) \geq x$, then $t \in A_x$, so $q = \inf A_x \leq t$.

($\Leftarrow$) We first show $F(q) \geq x$. For each $n \geq 1$, $q + n^{-1}$ is not a lower bound of A_x, so there is a $y_n \in A_x$ with $q \leq y_n < q + n^{-1}$. Then $y_n \to q$ with $y_n \geq q$, so $F(y_n) \to F(q)$ by the right continuity of F. Since $F(y_n) \geq x$ for all n, $F(q) \geq x$. Now suppose $q \leq t$. Since F is nondecreasing, $F(t) \geq F(q) \geq x$.

The cdf of Z_1. Let $t \in \mathbb{R}$ be given. By (1),

$$P(Z_1 \le t) = m\big(\{x \in (0,1) : F_1^{-1}(x) \le t\}\big) = m\big(\{x \in (0,1) : x \le F(t)\}\big) = F(t),$$

where the last equality holds because $0 \le F(t) \le 1$.

The cdf of Z_2. We claim that for any $x \in (0,1)$ and $t \in \mathbb{R}$,

$$F(t) > x \implies F_2^{-1}(x) \le t \implies F(t) \ge x. \tag{2}$$

For the first implication, suppose $F(t) > x$, and let $y \in B_x$. Then $F(y) \le x < F(t)$, so $y < t$ since F is nondecreasing. Thus t is an upper bound of B_x, and $F_2^{-1}(x) = \sup B_x \le t$.

For the second, suppose, for contradiction, that $F_2^{-1}(x) \le t$ but $F(t) < x$. By the right continuity of F, there is an $s > t$ with $F(s) < x$. Then $s \in B_x$, so $F_2^{-1}(x) \ge s > t$, a contradiction.

Now let $t \in \mathbb{R}$ be given. By (2),

$$\{x \in (0,1) : x < F(t)\} \subset \{x \in (0,1) : F_2^{-1}(x) \le t\} \subset \{x \in (0,1) : x \le F(t)\}.$$

The two outer sets differ by at most the point $F(t)$, and each has Lebesgue measure $F(t)$ since $0 \le F(t) \le 1$. (The middle set is then one of the two outer sets, and hence is Borel.) By the monotonicity of m,

$$P(Z_2 \le t) = m\big(\{x \in (0,1) : F_2^{-1}(x) \le t\}\big) = F(t).$$

Since t was arbitrary, Z_1 and Z_2 both have cdf F. //

(ii)

Show also that $F_1^{-1}(\cdot)$ is left continuous and $F_2^{-1}(\cdot)$ is right continuous.

Answer

Proof. Both functions are nondecreasing on $(0,1)$: for $0 < x \le x' < 1$, $A_{x'} \subset A_x$ and $B_x \subset B_{x'}$, so

$$F_1^{-1}(x) = \inf A_x \le \inf A_{x'} = F_1^{-1}(x') \quad \text{and} \quad F_2^{-1}(x) = \sup B_x \le \sup B_{x'} = F_2^{-1}(x').$$

Let $x \in (0,1)$ and $\epsilon > 0$ be given.

Left continuity of F_1^{-1}. Let $q \equiv F_1^{-1}(x)$. Since $q - \epsilon < \inf A_x$, $q - \epsilon \notin A_x$, i.e., $F(q-\epsilon) < x$. Set $\delta \equiv x - F(q-\epsilon) > 0$, and let $x' \in (x-\delta, x)$. Then $0 \le F(q-\epsilon) < x' < x < 1$, so $x' \in (0,1)$, and (1) applied to x' and $t = q - \epsilon$ gives $F_1^{-1}(x') > q - \epsilon$. Also $F_1^{-1}(x') \le q$, since F_1^{-1} is nondecreasing. Hence

$$0 \le F_1^{-1}(x) - F_1^{-1}(x') < \epsilon \quad \text{for all } x' \in (x-\delta, x).$$

Right continuity of F_2^{-1}. Let $s \equiv F_2^{-1}(x)$. Since $s + \epsilon > \sup B_x$, $s + \epsilon \notin B_x$, i.e., $F(s+\epsilon) > x$. Set $\delta \equiv F(s+\epsilon) - x > 0$, and let $x' \in (x, x+\delta)$. Then $0 < x < x' < F(s+\epsilon) \le 1$, so $x' \in (0,1)$, and the first implication in (2), applied to x' and $t = s + \epsilon$, gives $F_2^{-1}(x') \le s + \epsilon$. Also $F_2^{-1}(x') \ge s$, since F_2^{-1} is nondecreasing. Hence

$$0 \le F_2^{-1}(x') - F_2^{-1}(x) \le \epsilon \quad \text{for all } x' \in (x, x+\delta).$$

Since x and ϵ were arbitrary, F_1^{-1} is left continuous and F_2^{-1} is right continuous on $(0,1)$. //

Q2: 2.20

Apply Corollary 2.3.5 to show that for any collection $\{a_{ij} : i, j \in \mathbb{N}\}$ of nonnegative numbers,

$$\sum_{i=1}^{\infty}\left(\sum_{j=1}^{\infty} a_{ij}\right) = \sum_{j=1}^{\infty}\left(\sum_{i=1}^{\infty} a_{ij}\right).$$

Answer

Proof. Let $\Omega \equiv \mathbb{N}$, $\mathcal{F} \equiv \mathcal{P}(\mathbb{N})$, and let μ be the counting measure on $(\Omega, \mathcal{F})$. Every function on $\mathbb{N}$ is $\mathcal{F}$-measurable.

Lemma. If $h : \mathbb{N} \to [0, \infty]$, then $\int h\, d\mu = \sum_{k=1}^{\infty} h(k)$.

For $k \in \mathbb{N}$, let $g_k \equiv h(k) I_{\{k\}}$. Each g_k is a nonnegative simple function, so by (3.2 from Definition 2.3.2),

$$\int g_k\, d\mu = h(k)\, \mu(\{k\}) = h(k).$$

Also $h = \sum_{k=1}^{\infty} g_k$, since for each $j \in \mathbb{N}$, $g_j(j) = h(j)$ and $g_k(j) = 0$ for $k \neq j$.

By Corollary 2.3.5,

$$\int h\, d\mu = \sum_{k=1}^{\infty} \int g_k\, d\mu = \sum_{k=1}^{\infty} h(k).$$

Now for $i \in \mathbb{N}$, define $h_i : \mathbb{N} \to [0, \infty)$ by $h_i(j) \equiv a_{ij}$, and let $h \equiv \sum_{i=1}^{\infty} h_i$, so that $h(j) = \sum_{i=1}^{\infty} a_{ij} \in [0, \infty]$. By the lemma,

$$\int h_i\, d\mu = \sum_{j=1}^{\infty} a_{ij} \quad \text{and} \quad \int h\, d\mu = \sum_{j=1}^{\infty}\left(\sum_{i=1}^{\infty} a_{ij}\right).$$

Since each h_i is nonnegative and measurable, Corollary 2.3.5 gives

$$\sum_{j=1}^{\infty}\left(\sum_{i=1}^{\infty} a_{ij}\right) = \int \left(\sum_{i=1}^{\infty} h_i\right) d\mu = \sum_{i=1}^{\infty} \int h_i\, d\mu = \sum_{i=1}^{\infty}\left(\sum_{j=1}^{\infty} a_{ij}\right). \quad //$$

Q3: 2.25

If $f(x) = I_{Q_1}(x)$ where $Q_1 = \mathbb{Q} \cap [0, 1]$, $\mathbb{Q}$ being the set of all rationals, then show that for any partition P, $U(P, f) = 1$ and $L(P, f) = 0$.

Answer

Proof. Let $P = \{x_0, x_1, \ldots, x_n\}$ be a finite partition of $[0, 1]$, i.e., $0 = x_0 < x_1 < \cdots < x_n = 1$. As in Section 2.4, for $i = 0, 1, \ldots, n-1$, let

$$M_i = \sup\{f(x) : x_i \leq x \leq x_{i+1}\} \quad \text{and} \quad m_i = \inf\{f(x) : x_i \leq x \leq x_{i+1}\}.$$

Fix i. Since $x_i < x_{i+1}$, by the density of $\mathbb{Q}$ and of $\mathbb{R} \setminus \mathbb{Q}$ in $\mathbb{R}$, the interval $[x_i, x_{i+1}] \subset [0, 1]$ contains a rational r and an irrational s. Then $r \in Q_1$ and $s \notin Q_1$, so $f(r) = 1$ and $f(s) = 0$. Since $0 \leq f \leq 1$, it follows that $M_i = 1$ and $m_i = 0$.

By Definition 2.4.1,

$$U(P,f) = \sum_{i=0}^{n-1} M_i\,(x_{i+1} - x_i) = \sum_{i=0}^{n-1} (x_{i+1} - x_i) = x_n - x_0 = 1$$

and

$$L(P,f) = \sum_{i=0}^{n-1} m_i\,(x_{i+1} - x_i) = 0.$$

Since P was arbitrary, $U(P,f) = 1$ and $L(P,f) = 0$ for any partition P. //

Q4: Extra (A)

Let $f : \Omega \to \mathbb{R}$ be a $\langle \mathcal{F}, \mathcal{B}(\mathbb{R})\rangle$-measurable nonnegative function. Define, for $n \geq 1$ and $x \in \Omega$,

$$f_n(x) = \begin{cases} \dfrac{i-1}{2^n}, & \text{if } \dfrac{i-1}{2^n} \leq f(x) < \dfrac{i}{2^n} \text{ for } i = 1, \ldots, n2^n \\ n, & \text{if } f(x) \geq n. \end{cases}$$

Then prove the following:

(i)

For each $n \geq 1$, $f_n(x)$ is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R})\rangle$-measurable.

Answer

Proof. Let $n \geq 1$ be given, and let

$$A_{n,i} \equiv f^{-1}\Big(\Big[\tfrac{i-1}{2^n}, \tfrac{i}{2^n}\Big)\Big), \quad i = 1, \ldots, n2^n, \quad \text{and} \quad B_n \equiv f^{-1}\big([n, \infty)\big).$$

The intervals $\big[\frac{i-1}{2^n}, \frac{i}{2^n}\big)$, $i = 1, \ldots, n2^n$, and $[n, \infty)$ are disjoint with union $[0, \infty)$. Since $f \geq 0$, each $x \in \Omega$ lies in exactly one of $A_{n,1}, \ldots, A_{n,n2^n}, B_n$, it then follows that

$$f_n = \sum_{i=1}^{n2^n} \frac{i-1}{2^n}\, I_{A_{n,i}} + n\, I_{B_n}.$$

Each of these intervals is in $\mathcal{B}(\mathbb{R})$, and f is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R})\rangle$-measurable, so $A_{n,i} \in \mathcal{F}$ for all i and $B_n \in \mathcal{F}$ (see Definition 2.1.3).

Then for any $A \in \mathcal{F}$, I_A is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R})\rangle$-measurable, since $I_A^{-1}(B) \in \{\emptyset, A, A^c, \Omega\} \subset \mathcal{F}$ for all $B \in \mathcal{B}(\mathbb{R})$. Thus f_n is a finite linear combination of $\langle \mathcal{F}, \mathcal{B}(\mathbb{R})\rangle$-measurable functions, and hence is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R})\rangle$-measurable by Corollary 2.1.4. //

(ii)

For each $x \in \Omega$, $f_n(x)$ is increasing with n (i.e., $\{f_n(\cdot)\}$ is an increasing sequence of functions).

Answer

Proof. For $n \geq 1$, let $P_n \equiv \{j/2^n : j = 0, 1, \ldots, n2^n\}$. We first claim that for all $n \geq 1$ and $x \in \Omega$,

$$f_n(x) = \max\big\{p \in P_n : p \leq \min\{f(x), n\}\big\}. \tag{3}$$

There are two cases for us to then consider.

Case 1: $f(x) \geq n$. Then $\min\{f(x), n\} = n = \max P_n$, such that the right side of (3) is $n = f_n(x)$.

Case 2: $\frac{i-1}{2^n} \leq f(x) < \frac{i}{2^n}$ for some $i \in \{1, \ldots, n2^n\}$. Then $\min\{f(x), n\} = f(x)$, and $j/2^n \leq f(x)$ iff $j \leq i-1$, so the right side of (3) is $(i-1)/2^n = f_n(x)$.

In either case, (3) holds. Noting functional equivalence then, we can address the question posed. Let $x \in \Omega$ and $n \geq 1$ be given. Each $j/2^n \in P_n$ equals $2j/2^{n+1}$ with $2j \leq n2^{n+1}$, so $P_n \subset P_{n+1}$. Also $\min\{f(x), n\} \leq \min\{f(x), n+1\}$, so

$$\big\{p \in P_n : p \leq \min\{f(x), n\}\big\} \subset \big\{p \in P_{n+1} : p \leq \min\{f(x), n+1\}\big\},$$

such that $f_n(x) \leq f_{n+1}(x)$ by (3).

Since x and n were arbitrary, $f_n(x) \leq f_{n+1}(x)$ for all $x \in \Omega$ and $n \geq 1$. //

(iii)

Prove that $f_n(x) \to f(x)$ pointwise (i.e., for each fixed x), as $n \to \infty$.

Answer

Proof. Let $x \in \Omega$ be given. Since $f(x) \in [0, \infty)$, there exists an $n_0 \in \mathbb{N}$ such that $n_0 > f(x)$. For $n \geq n_0$, $0 \leq f(x) < n$, so $\frac{i-1}{2^n} \leq f(x) < \frac{i}{2^n}$ for some $i \in \{1, \ldots, n2^n\}$, and $f_n(x) = \frac{i-1}{2^n}$. Hence

$$0 \leq f(x) - f_n(x) < \frac{1}{2^n} \quad \text{for all } n \geq n_0,$$

and so $f_n(x) \to f(x)$ as $n \to \infty$.

Since x was arbitrary, $f_n \to f$ pointwise. //

Q5: Extra (B)

If $f, g \in L^1(\Omega, \mathcal{F}, \mu)$, show

$$\min(f, g) \in L^1(\Omega, \mathcal{F}, \mu)$$

and

$$\min\left(\int f\, d\mu, \int g\, d\mu\right) \geq \int \min(f, g)\, d\mu.$$

Answer

Proof. We first show $\min(f,g) \in L^1(\Omega, \mathcal{F}, \mu)$. For each $a \in \mathbb{R}$,

$$\{\omega : \min(f,g)(\omega) \leq a\} = \{\omega : f(\omega) \leq a\} \cup \{\omega : g(\omega) \leq a\} \in \mathcal{F},$$

so $\min(f,g)$ is $\mathcal{F}$-measurable (using Definition 2.1.2(a)).

At each $\omega \in \Omega$, $\min(f,g)(\omega)$ is either $f(\omega)$ or $g(\omega)$, so $|\min(f,g)| \leq |f| + |g|$. Then by Corollary 2.3.3,

$$\int |\min(f,g)|\, d\mu \leq \int \big(|f| + |g|\big)\, d\mu = \int |f|\, d\mu + \int |g|\, d\mu < \infty,$$

and so $\min(f,g) \in L^1(\Omega, \mathcal{F}, \mu)$.

We then turn to the inequality. Note that $\min(f,g) \leq f$ and $\min(f,g) \leq g$ on Ω. Since $\min(f,g), f, g \in L^1(\Omega, \mathcal{F}, \mu)$, Proposition 2.3.8 (ii) gives

$$\int \min(f,g)\, d\mu \leq \int f\, d\mu \quad \text{and} \quad \int \min(f,g)\, d\mu \leq \int g\, d\mu,$$

and so $\int \min(f,g)\, d\mu \leq \min\big(\int f\, d\mu, \int g\, d\mu\big)$. //